Asymmetric Information

Ques-01:- What are the informational asymmetries in market models? Examine with examples.

Answer:-

Informational Asymmetry:-

An informational asymmetry is present when one party to a transaction has more information than another about the characteristics of the good or service to be traded.

For example, the seller of a used car may know more about her car's quality than prospective buyers, or an applicant for an insurance policy may understand her health or disability risk better than the companies that offer such policies. Such asymmetries can exacerbate the ~~infla~~ inefficiencies that sometimes arise when information is imperfect. If buyers can't distinguish good cars from bad ones, sellers may be inclined to cars of very low quality. If insurance companies have difficulty evaluating applicants' risks, they may find themselves disproportionately serving high-risk policyholders.

There are two types of informational asymmetries in market models:-

(i) Adverse selection

(ii) Moral hazard

They are examined below.

Adverse Selection:-

Adverse selection is present if an informed individual is more willing to trade when trading is less advantageous to an uninformed trading partner. When uninformed parties realize that they face adverse selection, they may become reluctant to trade, causing markets to perform poorly.

Suppose there is a market of used cars. Because sellers want to sell lemons and keep good cars, buyers of used cars must be wary of quality. This consideration drives the price of a used car and reduces the number of good cars owners are willing to sell. In some cases, adverse selection can drive good cars owners from the market completely. Suppose all owners of lemons are willing to sell their cars regardless of the price, but owners of the good cars will become sellers only if the price of used cars is high enough. In this case, a reduction in price will raise the fraction of lemons among available used cars. This effect makes buyers even more wary of used cars and less willing to pay for them, which will drive the price further down. When the price falls, the fraction of lemons among the used-car supply will grow even larger. As a result of this vicious cycle, the price of a used car may be so low that no good cars are offered for sale.

Moral Hazard:-

Moral hazard is present when one party to a transaction takes actions that a trading partner cannot observe, and that affect the benefits the partner receives from the trade.

For example, automobile manufacturererers are free to vary the quality of inputs and care with which their cars are assembled. Customers are profoundly affected by those decisions, but rarely can observe them. Likewise, an insured individual's risk of death or disability may be high because of his behaviour - smoking, excessive drinking, or a lack of exercise. However, while the company from which he purchases life or disability insurance is obviously affected by those decisions, it is likely to have difficulty monitoring his behavior and adjusting its premium accordingly.

In a setting with moral hazard, the uninformed party wants to ensure that her trading partner takes action that promote her interests. Ideally she would write a contract that specifies the actions her partner must take. Unfortunately, if neither she nor a court can observe those actions, such a contract would be unenforceable.

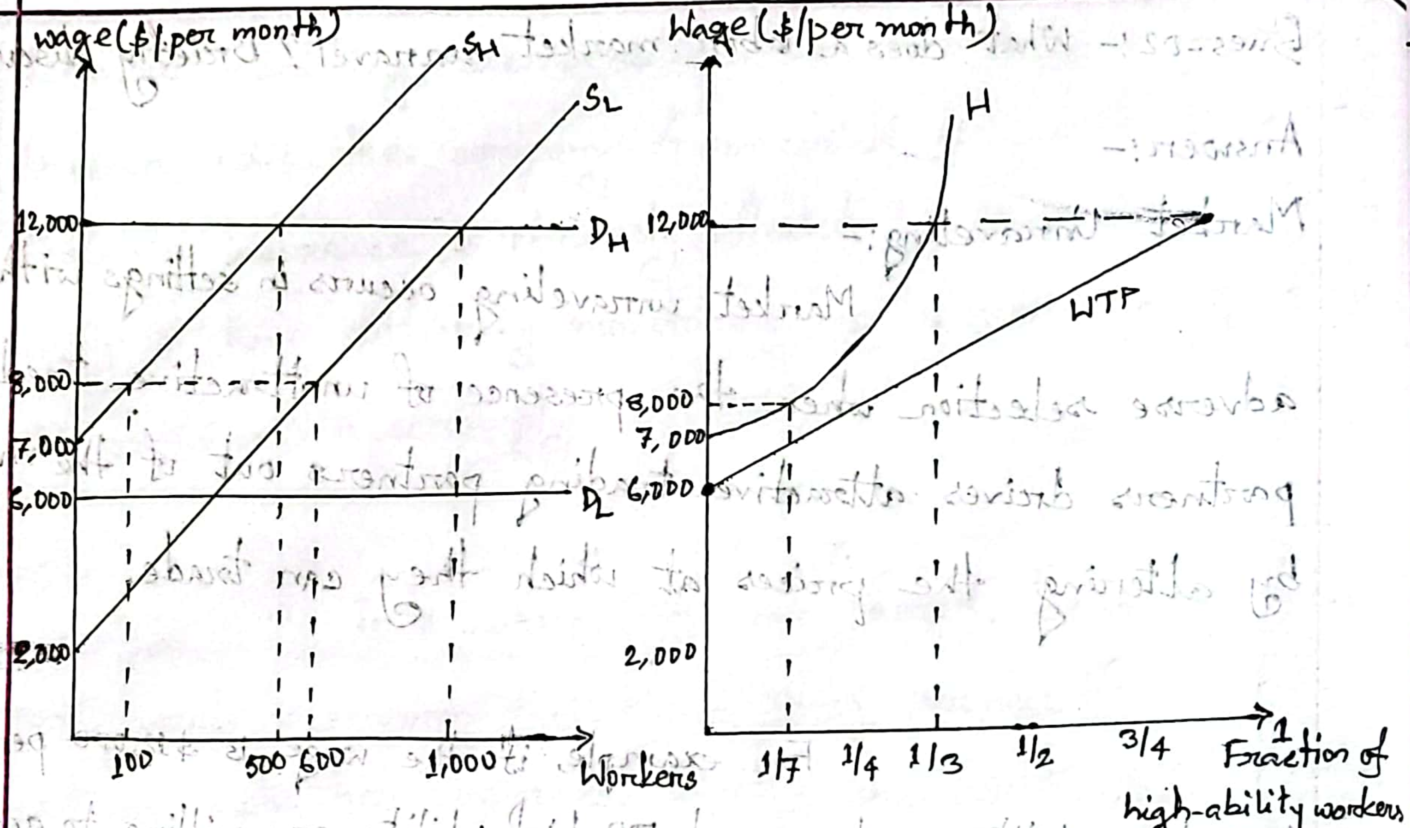
Ques-02:- What does a labor market unravel? Briefly discuss.

Answer:-

Market Unraveling:-

Market unraveling occurs in settings with adverse selection when the presence of unattractive trading partners drives attractive trading partners out of the market by altering the prices at which they can trade.

For example, if the wage is $12,000 per month, 1,000 low-ability workers and 500 high-ability are willing to accept employment. But when wage is $8,000 per month, 600 low-ability workers and 100 high-ability workers are willing to accept employment. And if wage falls below $7,000, only low-ability workers are willing to accept employment. In such cases, the presence of low-ability workers can chase high-ability workers out of the market entirely. This unfortunate outcome emerges even though employers would hire both high- and low-ability workers if information were perfect.



(a) Supply and demand

(b) Willingness to pay and the fraction of high-ability workers

Fig: Market unraveling due to Adverse selection

From fig-(a), there is no wage above $7,000 (the lowest wage at which high-ability workers are willing to accept employment) at which the demand for labor equals supply. Suppose the wage is $12,000. Then, two-third of the workers willing to accept employment have low-ability. The presence of these low-ability workers means that an employer won't be willing to pay more than $8,000. But if the wage is $8,000 or less, at least six-seventh of the workers willing to accept employment would have low-ability, so an employer would be unwilling to pay more than \$6,857 [since, \$6857 = $(1/7 \times 12{,}000) + (6/7 \times 6{,}000)$]. But with a wage less than \$7,000, no high-ability workers will seek employment. The equilibrium wage will be \$6,000 and employers will hire 400 low-ability workers. ~~So, asymmetric~~ Although perfectly informed employers would hire 500 high-ability workers, asymmetric information drives all of those workers out of the labor market.

Fig-(b) illustrates another approach to finding the market equilibrium. The height of the curve labeled WTP shows, for each such fraction, F_H, an employer's willingness to pay for workers: $WTP = (F_H \times 12{,}000) + [(1-F_H) \times 6{,}000]$. The curve H shows, for each possible wage the fraction of total labor that would be supplied by high-ability workers. The fraction is given by the formula:

$$F_H = \frac{S_H(W)}{S_L(W) + S_H(W)}$$

where $S_H(W)$ and $S_L(W)$ are the supplies of high- and low-ability workers, respectively, at a wage of W. An equilibrium occurs at the intersection of these two curves. In such an equilibrium the market wage W equals the average productivity of those workers seeking employment given that wage; employers are therefore willing to hire all of the workers who seek employment.

In fig-(b), the curves intersect where the fraction of high-ability workers is zero; therefore, employers hire no high-ability workers.

Ques-03:- Suppose each entry-level software programmer in Palo Alto, California, has either high or low ability. All potential employers value a high-ability worker at \$12,000 per month and a low-ability worker at \$6,000. The supply of high-ability workers is $Q_H = 0.05(W-2000)$ and the supply of low-ability workers is $Q_L = 0.1(W-2000)$, where W is the monthly wage. If workers' abilities are observable to employers, what are the equilibrium wages? How many workers of each type do employers hire? If workers' abilities are not observed by employers, what is the equilibrium wage? How many workers of each type do employers hire? What is the deadweight loss due to asymmetric information? Graphically present your findings.

Answer:-

When workers' abilities are observable, the wage for a high-ability worker must equal \$12,000, his value to employers. We can calculate the number hired from the supply function:

$Q_H = 0.05(12,000 - 2,000) = 500.$

In a similar ~~way~~ fashion, we find that the wage of a low-ability worker is \$6,000 and the number hired is $Q_L = 0.1(6,000 - 2,000) = 400$.

When workers' ability are not observed by employers, there is a single wage, W. The aggregate labor supply function is

$$Q^S = Q^S_H + Q^S_L$$

$$= 0.05(W - 2,000) + 0.1(W - 2,000)$$

$$= 0.15(W - 2,000)$$

An employer's willingness to pay depends on the fraction of available workers who have high-ability, F_H. That fraction equals

$$F_H = \frac{Q^S_H}{Q^S_H + Q^S_L} = \frac{0.05(W - 2,000)}{0.05(W - 2,000) + 0.1(W - 2,000)} = \frac{0.05}{0.15} = \frac{1}{3}$$

Notice that this fraction is the same at every possible wage. So the expected value of a job applicant to an employer is

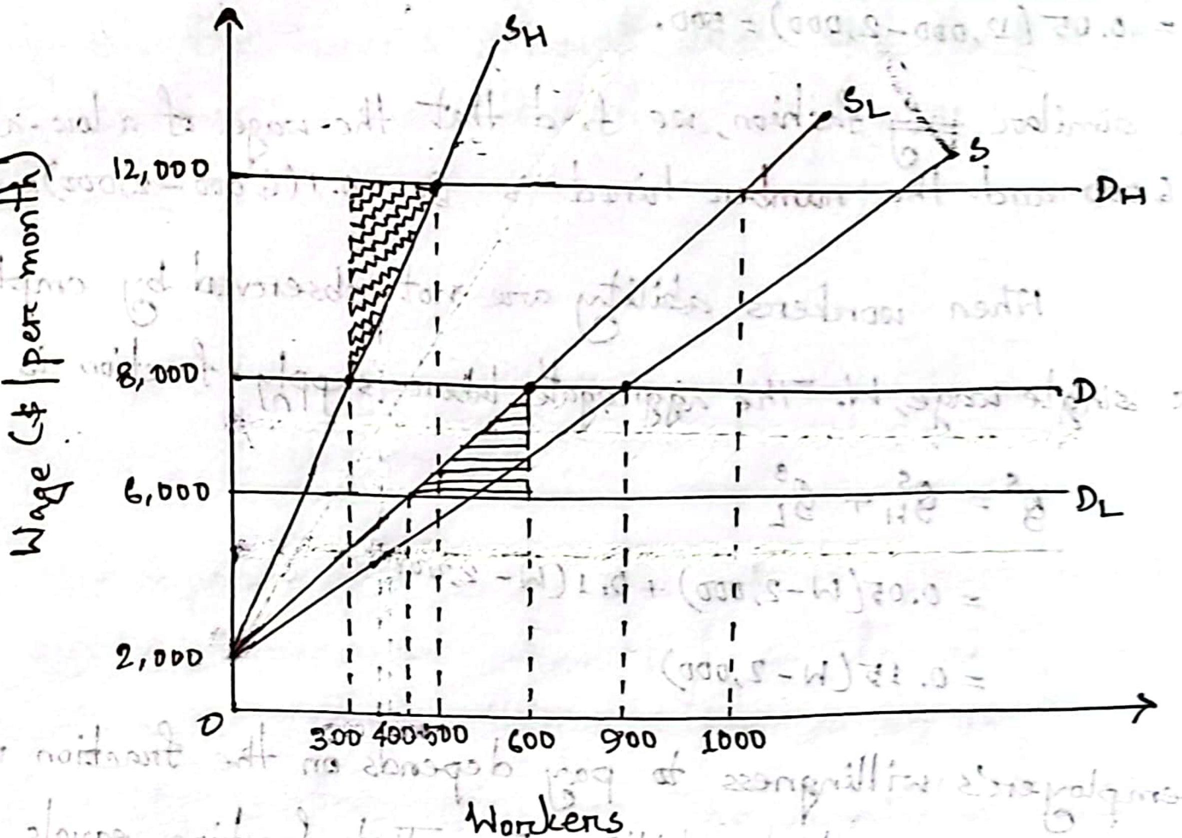
$$= \left[(1/3) \times 12,000 + (2/3) \times 6,000\right]$$

= \$8,000 regardless of the wage. The equilibrium wage is therefore \$8,000.

We calculate the total number of workers hired from the supply function: $Q^S = 0.15(8,000 - 2,000) = 900.$

Number of high-ability workers $= \frac{1}{3} \times 900 = 300$

Number of low-ability workers $= 900 - 300 = 600$



Employers hire too many low-ability workers (600 instead of 400) and too few high-ability workers (300 instead of 500). The deadweight loss from hiring too many low-ability workers is the straight-shaded area between demand and supply curve D_L and S_L in the figure, which equals,

$$= \frac{1}{2} \times (200 \times 2{,}000)$$

$$= \$2{,}00{,}000 \text{ per month}$$

The deadweight loss from hiring too few high-ability workers is the curved-shaded area between the demand and supply curve D_H and S_H in the figure, which equals

$$= \frac{1}{2} \times (200 \times 4{,}000)$$

$$= \$4{,}00{,}000 \text{ per month}$$

So total deadweight loss is \$2,00,000 + \$4,00,000 = \$6,00,000 per month.

All the findings are presented graphically above in the figure.

Ques-04:- Each entry-level software programmer in Palo Alto, California, has either high or low ability. All potential employers value a high-ability worker at \$12,000 per month and a low-ability worker at \$6,000. The supply of high-ability worker is $Q^S_H = 0.05(W - 2,000)$ and the supply of low-ability worker is $Q^S_L = 0.15(W - 2,000)$, where W is the monthly wage. If workers' abilities are observable to employers, what are the equilibrium wages? How many workers of each type do employers hire? If workers' abilities are not observed by employers, what is the equilibrium wage? How many workers of each type do employers hire? What is the deadweight loss due to asymmetric information?

Answer:-

When workers' abilities are observable, the wage for a high-ability worker must equal \$12,000, his value to employers. We can calculate the number hired from the supply function:

$$Q_H = 0.05(12,000 - 2,000)$$

$$= 500.$$

In a similar fashion, we find that the wage of a low-ability worker is \$6,000 and the number hired is

$$S_L = 0.15(6,000-2,000)$$
$$= 600$$

When workers' abilities are not observed by employers, there is a single wage, W. The aggregate labor supply function is

$$S^S = S_H^S + S_L^S$$
$$= 0.05(W-2,000) + 0.15(W-2,000)$$
$$= 0.20(W-2,000)$$

An employer's willingness to pay depends on the fraction of available workers who have high-ability, F_H. That fraction equals

$$F_H = \frac{S_H^S}{S_H^S + S_L^S} = \frac{0.05(W-2,000)}{0.05(W-2,000)+0.15(W-2,000)} = \frac{0.05}{0.20} = \frac{1}{4}$$

Notice that this fraction is the same at every possible wage. So the expected value of a job applicant to an employer is

$$= [(1/4)\times(\$12,000) + (3/4)\times \$6,000]$$

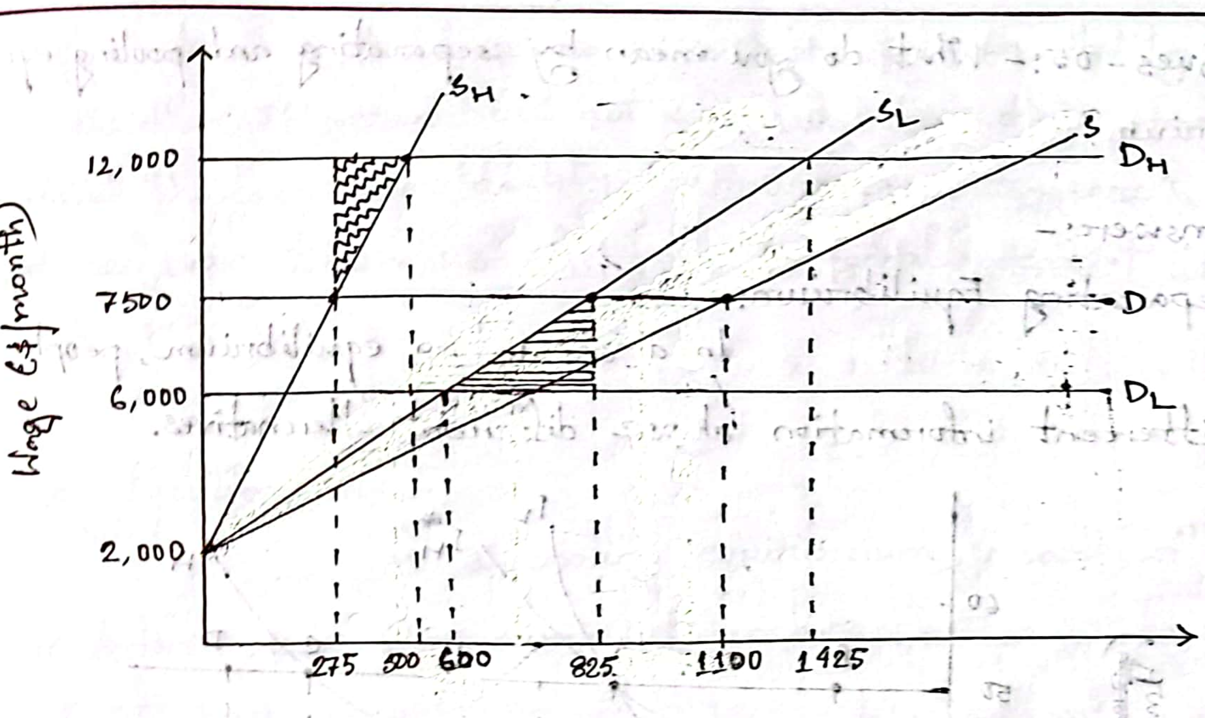
$= \$7,500$ regardless of the wage. The equilibrium wage is therefore \$7,500.

We calculate the total number of workers hired from the supply function: $S^S = 0.20(7500-2000)$

$$= 1100$$

The number of high-ability workers hired $= \frac{1}{4}\times 1100 = 275$

The number of low-ability workers hired $= (1100-275) = 825.$



Employers hire too many low-ability works (825 instead of 600) and too few high-ability workers (275 instead of 500). The deadweight loss from hiring too many low-ability workers is straigth-shaded area between the demand and supply curves D_L and S_L in the figure, which equals $= \frac{1}{2}(225 \times 1500)$

$= \$1,68,750$ per month.

The deadweight loss from hiring too ~~man~~ few high-ability workers is curved-shaded area between the demand and supply curves D_H and S_H in the figure which equals $= \frac{1}{2}(225 \times 4500)$

$= \$5,06,250$ per month.

So total deadweight loss is $= \$1,68,750 + \$5,06,250$

$= \$6,75,000$ per month.

Ques-05:- What do you mean by separating and pooling equilibrium?

Answer:-

Separating Equilibrium:-

In a separating equilibrium, people with different information choose different alternatives.

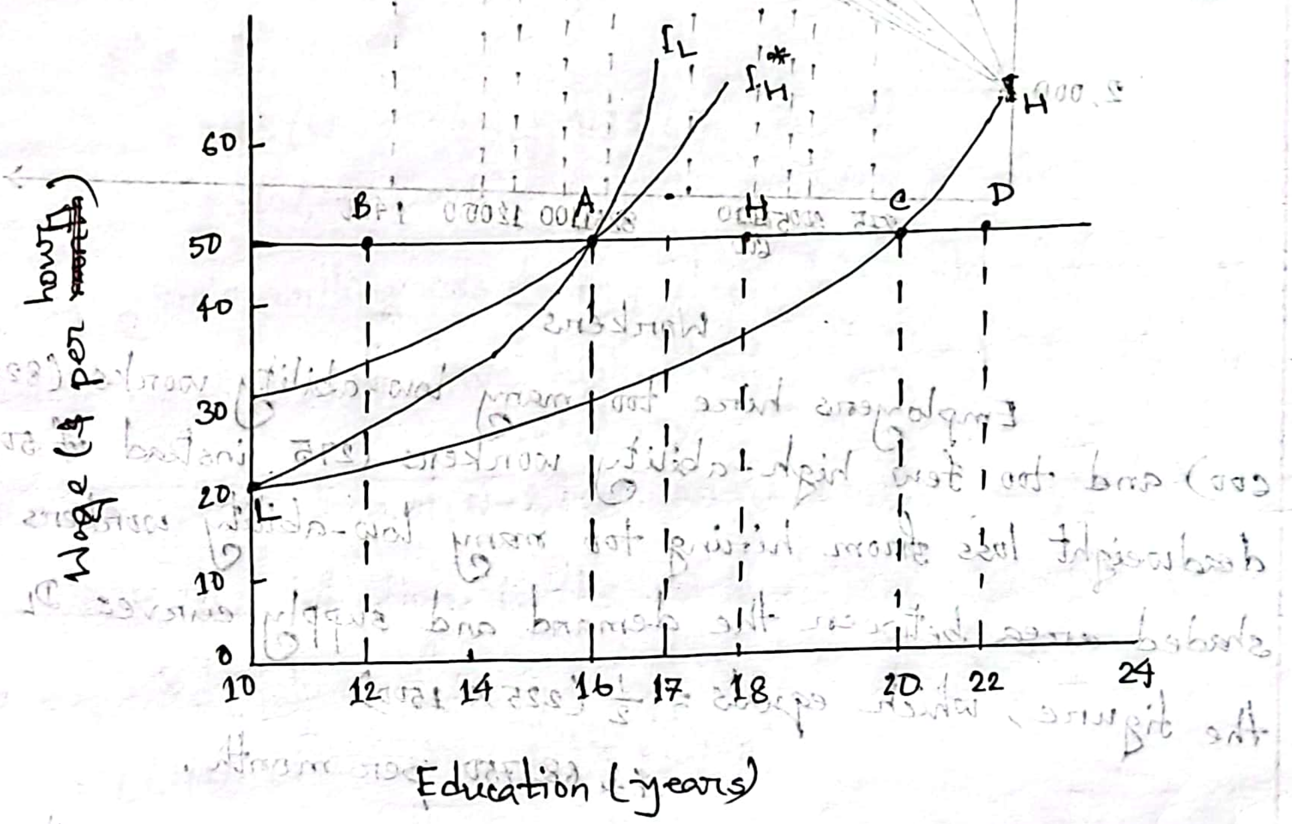


Fig: Equilibrium with Separation

In a separating equilibrium, low-ability workers end up at point L. They obtain the minimum amount of education required by law (10 years) and are paid the value of their marginal product ($20 per hour). High-ability workers end up at a point such as H, between point A and C. They obtain enough education to discourage imitation by low-ability workers (no less than 16 years), but not so much that they want to imitate those with low ability (no more than 20 years). They too ~~paid~~ are paid the value of their marginal product ($50 per hour).

Pooling Equilibrium:-

In a pooling equilibrium, people with different information choose the same alternative.

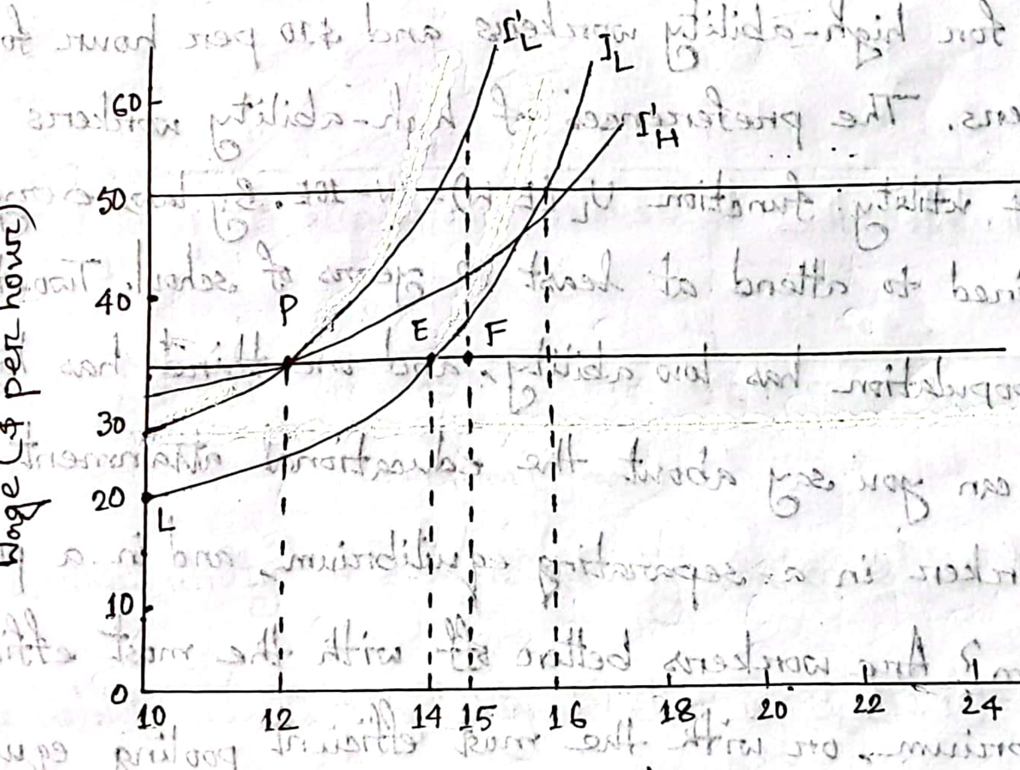


Fig: Equilibrium with pooling

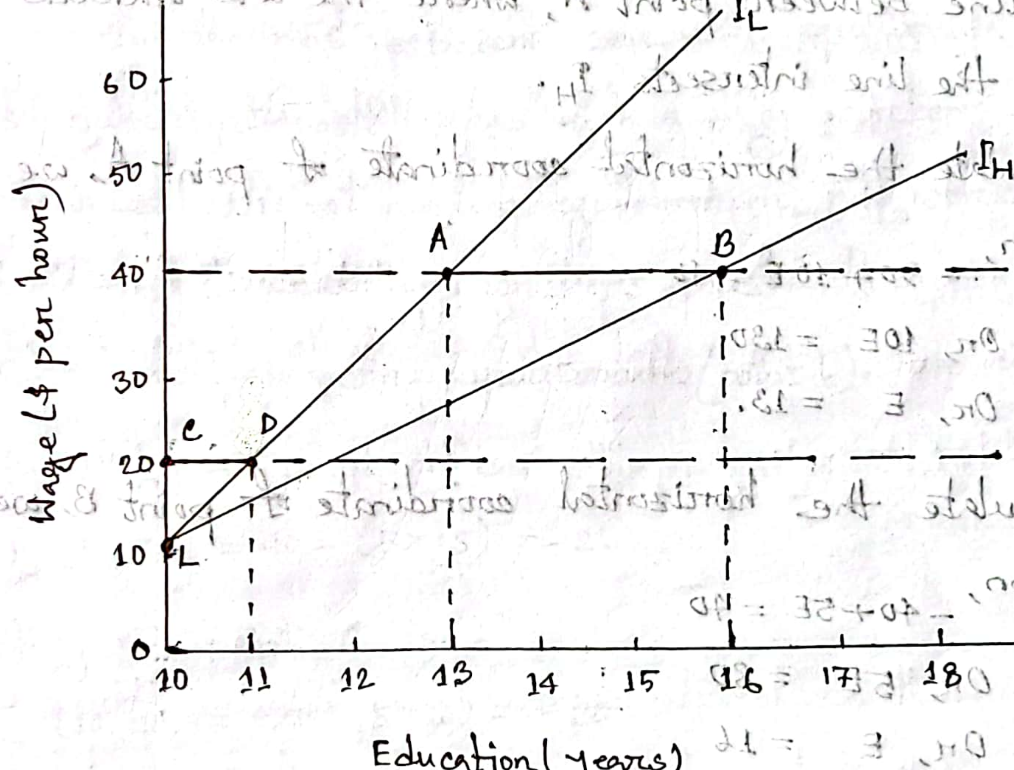
In a pooling equilibrium, high- and low-ability workers receive the same amount of education and are paid the expected value of the marginal product for a randomly selected worker ($35 per hour). Workers must not obtain so much education that those with low ability would instead choose to obtain the minimum level of education and receive $20 per hour.

Therefore, workers end up at a point such as P, between vertical axis and point E.

Ques-06:- The value of worker's marginal product is $40 per hour for high-ability workers and $10 per hour for low-ability workers. The preferences of high-ability workers correspond to the utility function $U_L(E, W) = W - 10E$. By law, everyone is required to attend at least 10 years of school. Two-thirds of the population has low ability, and one-third has high ability. What can you say about the educational attainment of each type of worker in a separating equilibrium, and in a pooling equilibrium? Are workers better off with the most efficient separating equilibrium, or with the most efficient pooling equilibrium?

Answer:-

We'll start with separating equilibrium. ~~For the same~~ Firms will pay $40 per hour (the value of the marginal product created by a high-ability worker) to those with E_H years of schooling, and \$10 per hour (the value of the marginal product created by a low-ability worker) to those with E_L years of schooling. Also, low-ability workers will obtain only the minimum amount of education required by law ($E_L = 10$). Thus, low-ability workers end up at point labeled L in figure.



At point L, the utility of a high-ability worker is $U_H(10, 10) = 10 - (5 \times 10) = -40$

while, the utility of a low-ability worker is $U_L(10, 10) = 10 - (10 \times 10) = -90$.

We therefore obtain the following formulas for indifference curves through point L:

$W = -40 + 5E$ for high-ability workers

$W = -90 + 10E$ for low-ability workers

We're drawn those indifference curves in the figure. The one labeled I_H belongs to a high-ability worker; and the one labeled I_L belongs to a low-ability worker. We have also drawn a horizontal line that intersects the vertical axis at \$40. High-ability workers must end up at a point on the thick segment of that line between point A, where the line intersects I_L, and B where the line intersects I_H.

To calculate the horizontal coordinate of point A, we solve the equation,

$$-90+10E=40$$
$$\text{Or, } 10E = 130$$
$$\text{Or, } E = 13.$$

To calculate the horizontal coordinate of point B, we solve the equation,

$$-40+5E=40$$
$$\text{Or, } 5E = 80$$
$$\text{Or, } E = 16$$

Therefore, E_H can be any value between 13 and 16.

Now we turn to pooling equilibria. The marginal product of an average worker is $(2/3 \times 10)+(1/3 \times 40) =\$20$ per hour. We have drawn a horizontal line in the figure that intersects the vertical axis at \$20 (point C). The indifference curve labeled I_L intersects that line at point D. All workers must end up on the thick segment of the line between points C and D. To calculate the horizontal coordinate of point D, we solve the equation

$$-90+10E=20$$

$$\text{Or, } 10E=110$$

$$\text{Or, } E=11$$

Therefore, E_p can be any value between 10 and 11.

In the most efficient separating equilibrium, low-ability workers end up at point L; and high-ability workers end up at point A. In the most efficient pooling equilibrium, all workers end up at point C. Clearly, low-ability workers prefer the pooling equilibrium (point C) to the separating equilibrium (point L). For high-ability workers, utility at point A is

$$U_H(13,40)=40-(5\times13)=-25$$

and utility at point C is

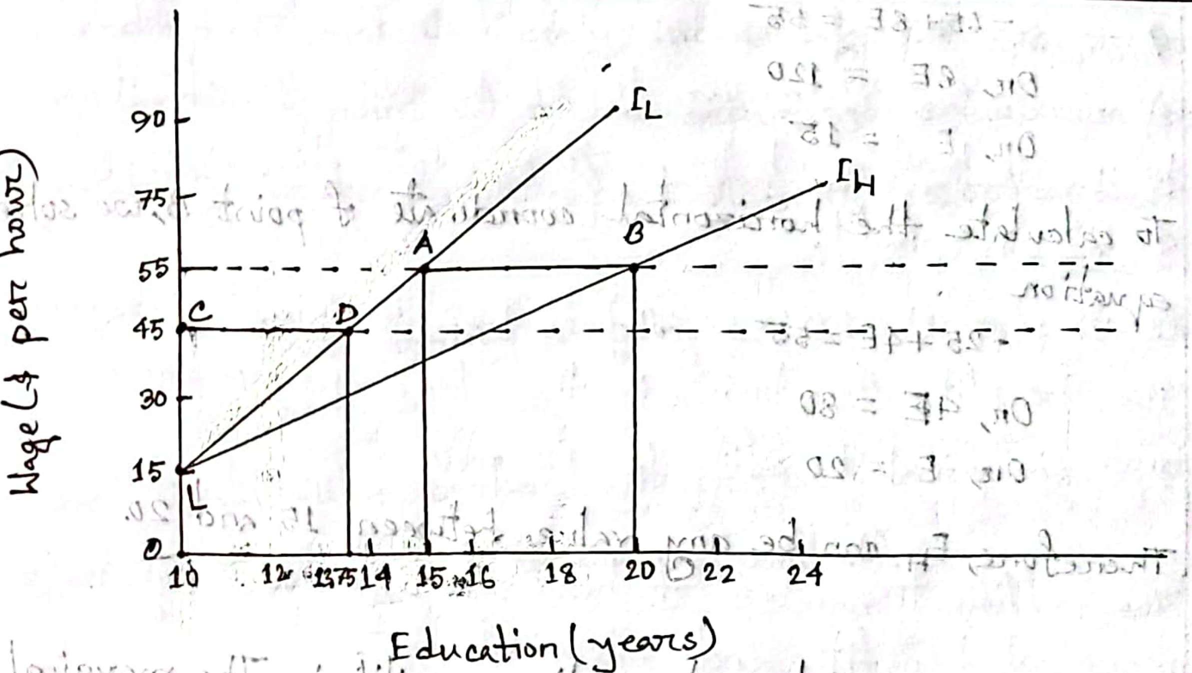
$$U_H(10,20)=20-(5\times10)=-30$$

Therefore, high-ability workers prefer the separating equilibrium (point A) to the pooling equilibrium (point C).

Ques-07:- The value of a worker's marginal product is \$55 per hour for high-ability workers and \$15 per hour for low-ability workers. The preferences of high-ability workers correspond to the utility function $U_H(E,W) = W - 4E$, and the preferences of low-ability workers correspond to the utility function $U_L(E,W) = W - 8E$. By law, everyone is required to attend at least 10 years of school. One-quarter of the population has low-ability, and three-quarters have high-ability. What can you say about the educational attainment of each type of worker in a separating equilibrium, and in a pooling equilibrium? Are workers better off with the most efficient separating equilibrium, or with the most efficient pooling equilibrium?

Answer:-

We'll start with separating equilibrium. Firms will pay \$55 per hour (the value of the marginal product created by a high-ability worker) to those with E_H years of schooling, and \$15 per hour (the value of the marginal product created by a low-ability worker) to those with E_L years of schooling. Also, low-ability workers will obtain only the minimum amount of education required by law ($E_L = 10$). Thus, low-ability workers end up at the point labeled L in the figure.



At point L,

the utility of a high-ability worker is $U_H(10, 15) = 15-(4\times10) = -25$

while the utility of a low-ability worker is $U_L(10, 15) = 15-(8\times10) = -65$.

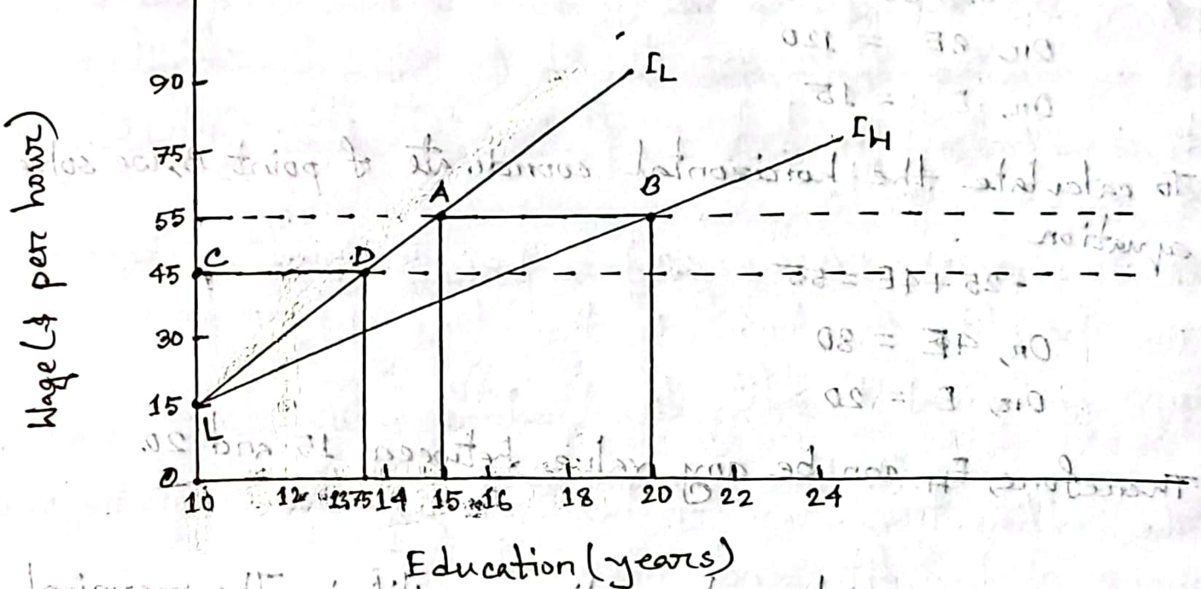
We therefore obtain the following formulas for indifference curves through point L:

$W = -25+4E$ for high-ability workers

and $W = -65+8E$ for low-ability workers.

We've drawn these indifference curves in the figure. The one labeled I_H belongs to a high-ability worker; and the one labeled I_L belongs to a low-ability worker. We've also drawn a horizontal line that intersects the vertical axis at \$55. High-ability workers must end up at a point on the thick segment of that line between point A, where the line intersects I_L, and B, where the line intersects I_H.

To calculate the horizontal coordinate of point A, we solve the equation.



At point L,

the utility of a high-ability worker is $U_H(10, 15) = 15 - (4 \times 10) = -25$

while the utility of a low-ability worker is $U_L(10, 15) = 15 - (8 \times 10) = -65$.

We therefore obtain the following formulas for indifference curves through point L:

$W = -25 + 4E$ for high-ability workers

and $W = -65 + 8E$ for low-ability workers.

We've drawn these indifference curves in the figure. The one labeled I_H belongs to a high-ability worker; and the one labeled I_L belongs to a low-ability worker. We've also drawn a horizontal line that intersects the vertical axis at \$55. High-ability workers must end up at a point on the thick segment of that line between point A, where the line intersects I_L, and B, where the line intersects I_H.

To calculate the horizontal coordinate of point A, we solve the equation

$$-65+8E = 55$$

Or, $8E = 120$

Or, $E = 15$

To calculate the horizontal coordinate of point B, we solve the equation

$$-25+4E = 55$$

Or, $4E = 80$

Or, $E = 20$

Therefore, E_H can be any value between 15 and 20.

Now we turn to pooling equilibria. The marginal product of an average worker is $(1/4 \times \$15)+(3/4 \times \$55) = \$45$ per hour. We've drawn a horizontal line in the figure that intersects vertical axis ~~at po~~ at \$45 (point C). The indifference curve labeled I_L intersects that line at point D. All workers must end up on the thick segment of that line between points C and D. To calculate the horizontal coordinate of point D, we solve the equation

$$-65+8E = 45$$

Or, $8E = 110$

Or, $E = 13.75$

Therefore, E_P can be any value between 10 and 13.75.

In the most efficient separating equilibrium, low-ability workers end up at point L, and high-ability workers end up at point A. In most efficient pooling equilibrium, all workers end up at point C. Clearly, low-ability workers prefer the pooling equilibrium (point C) to the separating equilibrium (point L).

For high-ability workers, utility at point A is $U_H(15, 55) = 55 - (4 \times 15)$
$= -5$

and utility at point C is $U_H(10, 45) = 45 - (4 \times 10)$
$= 5$

Therefore, high-ability workers also prefer the pooling equilibrium (point C) to the separating equilibrium (point A).

Ques-08:- A salesperson works for a car dealership for 40 hours per week, but may not choose to work hard all of the time. The dealership's owner cannot observe the salesperson's effort, but can observe the number of cars sold. The salesperson's personal cost of working at the dealership is $C = 1,000 + H^2$, where H is the number of hours during which he works hard. The corresponding marginal cost is $MC = 2H$. Without any effort, the salesperson will, on average, generate a profit of \$800. With each hour of high effort, he has a 4 percent chance of selling a car. Each car sale generates a profit of \$1000. What is the efficient number of hours of high effort? How much surplus does the relationship generate?

Answer:-

We first derive the dealership's benefit function. Each hour of hard work has a 4 percent chance of generating \$1,000 in profit.

So it produces (\$1000 × 0.04) or, \$40 in profit on average.

The benefit function is therefore $B = 800 + 40H$

The marginal benefit is $MB = 40$.

The efficient number of hours of high effort equates the marginal benefit and marginal cost.

Setting $MB = MC$

Or, $40 = 2H$

Or, $H = 20$

So the efficient number of hours of high effort = 20 hours.

~~And net benefit, B = 800 + (40)(20)~~

And surplus = Benefit − Cost

$= [800 + (40)(20)] - [1000 + (20)^2]$

$= 1600 - 1400$

$= \$200$

Second part of 8:- Suppose the owner gives the salesperson \$1,200 in base pay plus a bonus of \$500 for each car he sells. How hard will the salesperson work? Describe an incentive scheme that leads to the efficient effort level and allows the owner to keep all of the surplus.

Answer:-

If the salesperson faces an incentive scheme with a base pay of \$1,200 and a bonus of \$500 for each car he sells, then each hour of high effort yeilds, on average, (\$500 × 0.04) or \$20 in extra compensation.

The salesperson's expected earnings are, $E = 1,200 + 20H$

and his marginal earnings (marginal benefit) are, $ME = 20$.

His best choice equates his marginal benefit and marginal cost.

Setting, $ME = MC$

Or, $20 = 2H$

Or, $H = 10$

And surplus = Benefit - Cost

$= E - C$

$= [1200 + 20H] - [1,000 + H^2]$

$= [1200 + (20)(10)] - [1000 + (10)^2]$

$= 1400 - 1100$

$= \$300$

So, the salesperson works hard for 10 hours and ~~earns~~ generates \$300 surplus.

For the salesperson to choose the efficient number of hours, he must receive all the benefits and absorb all the costs of his actions on the margin. Thus, his marginal benefit at 20 hours must be \$40. Consider a linear incentive scheme with base pay of K and a bonus of \$1,000 per car (the full profit). In that case, the salesperson's marginal benefit from an hour of hard work will be \$40 on average. To ensure that the owner keeps all of the surplus, we set K so that the salesperson's net benefit is zero (his total earnings equals total cost):

$$K + (40)(20) = 1000 + (20)^2$$

$$\text{Or, } K + 800 = 1400$$

$$\text{Or, } K = 600$$

So the owner offers the salesperson a contract that gives him a base pay of \$600 and pays him a bonus of \$1,000 for each car he sells. With this incentive scheme, the owner retains all of the surplus.

Ques-09:- What do you mean by an incentive scheme? Suppose a salesperson works for a car dealership for 40 hours per week, but may not choose to work hard all of the time. The dealership's owner cannot observe the salesperson's effort, but can observe the number of cars sold. The salesperson's personal cost of working at the dealership is $C = 1{,}000 + H^2$, where H is the number of hours during which he works hard. The corresponding marginal cost is $MC = 2H$. Without any effort, the salesperson will on average, generate a profit of \$800. With each hour of high effort, he has 10 percent chance of selling a car. Each car sale generates a profit of \$1000. What is the efficient number of hours of high effort? How much surplus does the relationship generate?

Answer:-

Incentive Scheme:-

An incentive scheme is a contract or compensation policy that ties rewords or punishments to performance, designed in a manner to induce desirable behavior.

Many examples of incentive schemes are observed in professional sports. Baseball players' contracts often include incentives tied to measurable aspects of individual and team-oriented performance. These may include bonuses for batting over 0.3000 (30 percent) and for reaching the playoffs. Such performance measures do not perfectly capture the amount of effort the player expended to ensure success. Instead, the player's pay is conditioned on outcomes that are related to his effort, but that are also affected by factors outside of his control, such as the abilities of other teams and luck. Nonetheless, tying pay to these outcomes creates incentives for the player to exert effort.

Solution:-

We first derive the dealership's benefit function. Each hour of hard work has a 10 percent chance of generating \$1,000 in profit.

So it produces (\$1000×.10) or, \$100 profit on average.

The benefit function is therefore $B = 800 + 100H$

and the marginal benefit is $MB = 100$.

The efficient number of hours of high ~~profit~~ effort equates the marginal benefit and marginal cost.

Setting $MB = MC$

Or, $100 = 2H$

Or, $H = 50$

$\therefore$ The efficient number of hours of high effort $= 50$.

Net surplus = Benefit - Cost

$= [800 + (100)(50)] - [1000 + (50)^2]$

$= 5800 - 3500$

$= \$2300$

$\therefore$ The relationship generates a surplus of \$2300.

Ques-10:- Suppose the owner gives the salesperson \$1,200 in base pay plus a bonus of \$500 for each car he sells. How hard will the salesperson work? Graphically describe an incentive scheme that leads to the efficient effort level and allow the owner to keep all the surplus.

Answer:-

If the salesperson faces an incentive scheme with a base pay of \$1,200 and a bonus of \$500 for each car he sells, then each hour of high effort yeilds, on average, (500×0.10) or, \$50 in extra compensation. The salesperson's expected earnings are $E = 1,200 + 50H$

and his marginal earnings (marginal benefits) are $ME = 50$.

His best choice equates this marginal benefit and marginal cost.

Setting $ME = MC$

Or, $50 = 2H$

Or, $H = 25$

His net surplus is $= E - C$

$= [1,200 + (50)(25)] - [1,000 + (25)^2]$

$= 2450 - 1625$

$= \$825$

So the salesperson will work hard for 25 hours and generate a surplus of $825.

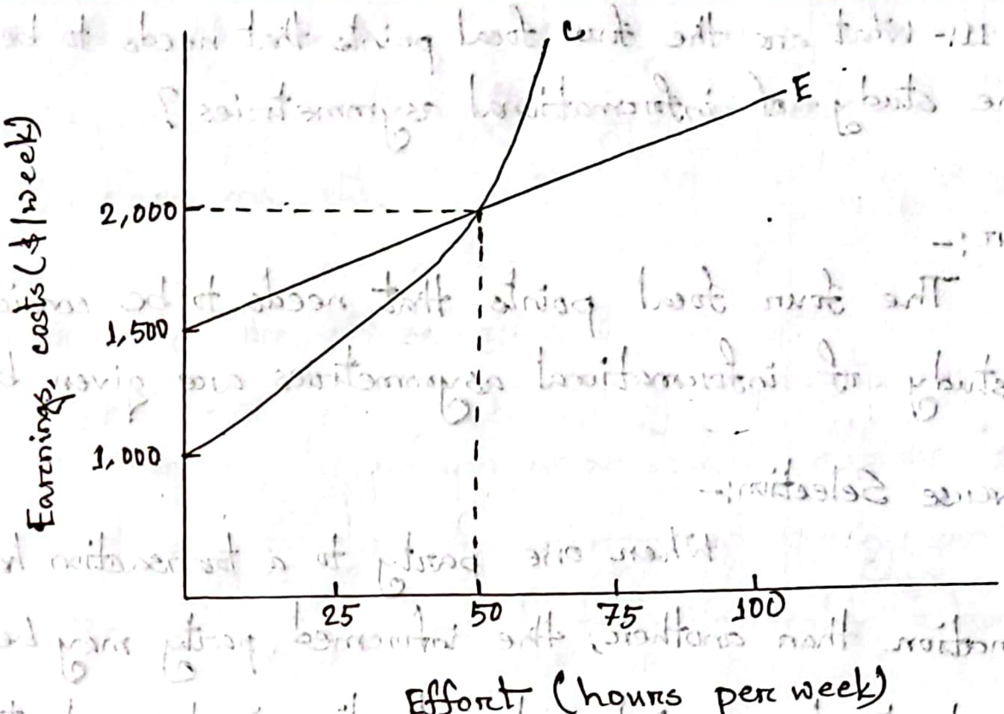
For the salesperson to choose the efficient number of hours, he must receive all the benefits and absorb all the costs of his actions on the margin. Thus, his marginal benefit at 50 hours must be $100. Consider a linear incentive scheme with base pay of K and a bonus of $1,000 per car (the full profit). In that case, the salesperson's marginal benefit from an hour of hard work will be $100 on average. To ensure that the owner keeps all of the surplus, we set K so that the salesperson's net benefit is zero (his total earnings equals his total cost):

$$K + (100)(50) = 1,000 + (50)^2$$

$$\text{Or, } K + 5000 = 3500$$

$$\text{Or, } K = 1500$$

While offering the salesperson the full profit from all car sales induces him to make an efficient choice, it leaves the owner with no profit; the salesperson receives all the surplus that their relationship creates. To achieve an



Figure

efficient outcome, however the owner only needs to provide the salesperson with the right incentives on the margin. In figure, if the owner gives the salesperson a base pay of $1,500 plus a bonus of $1,000 for each car he ~~sales~~ sells, the salesperson will work hard for 50 hours and receive total compensation equal to his personal costs (including opportunity costs). With this incentive scheme, the owner receives all of the surplus.

Ques-11:- What are the four focal points that needs to be considered in the study of informational asymmetries?

Answer:-

The four focal points that needs to be considered in the study of informational asymmetries are given below:-

(i) Adverse Selection:-

When one party to a transaction has more information than another, the informed party may be more willing to trade precisely when trading is less advantageous to the uninforomed party. As a result, the uninformed party may be reluctant to trade. This reluctance can cause markets to perforum poorly.

(ii) Signaling:-

Informed individuals often undertake costly activities to convince others of particular facts. Such actions can convey information and create effect on the efficiency of resource allocation.

(iii) Screening:-

Faced with an informational handicap, an uninformed party may establish a test that induces informed parties to self-select, thereby revealing what they know. Government can in principle improve upon the efficiency, fairness, and stability of screening in free markets.

(iv) Incentives and Moral Hazard:-

In many circumstances, the attributes of a good or service depend on unobservable actions taken by one or more of the trading parties. An uninformed party can provide a trading partner with incentives to take favorable actions.

Ques-12:- 'Due to the informational asymmetry low ability workers can drive high ability workers from the market either partially or completely' justify the statement with necessary diagrams and logic.

Answer:-

Suppose there are two types of workers – low-ability and high-ability. A high-ability worker generates \$12,000 of profit per month, while a low-ability worker generates only \$6,000. Fig-1 shows the demand curves for high- and low-ability workers when workers' abilities are observable to employers, labeled D_H and D_L respectively. The figure also shows the supply curve of high- and low-ability workers labeled S_H and S_L respectively. The higher the wage, the more high-ability workers are willing to accept employment.

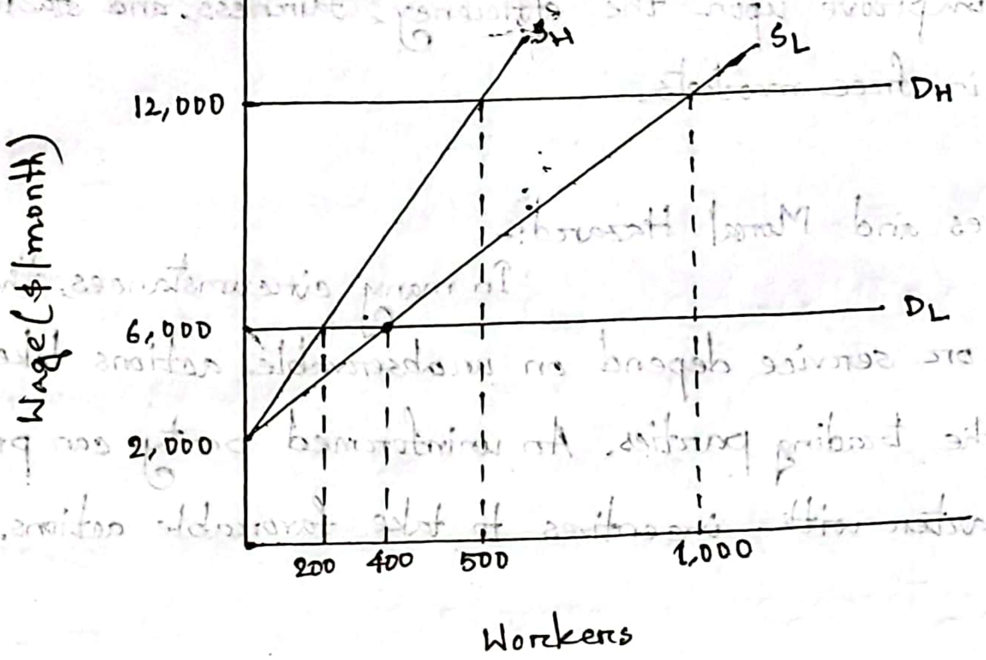


Fig-1: Demand and Supply of labor when workers' abilities are observable

When a worker's ability is perfectly observable, high- and low-ability workers receive different wages. In a perfectly competitive labor market, the wages of high- and low-ability workers adjust so that the quantity demanded equals the quantity supplied of each type. In fig-1, employers hire 400 low-ability workers at a monthly wage of $6,000 and 500 high-ability workers at a monthly wage of $12,000.

Now suppose that firms can't tell whether a worker has high or low ability. If employers cannot distinguish between high- and low-ability workers, then they must pay all workers the same wage, which will be between the values of the high- and low-ability workers. Employers will pay high-ability workers less than their value and low-ability workers more than their value. As a result, low-ability workers will be more likely and high-ability workers less likely to accept employment. Thus, a larger fraction of the available labor supply will have low ability.

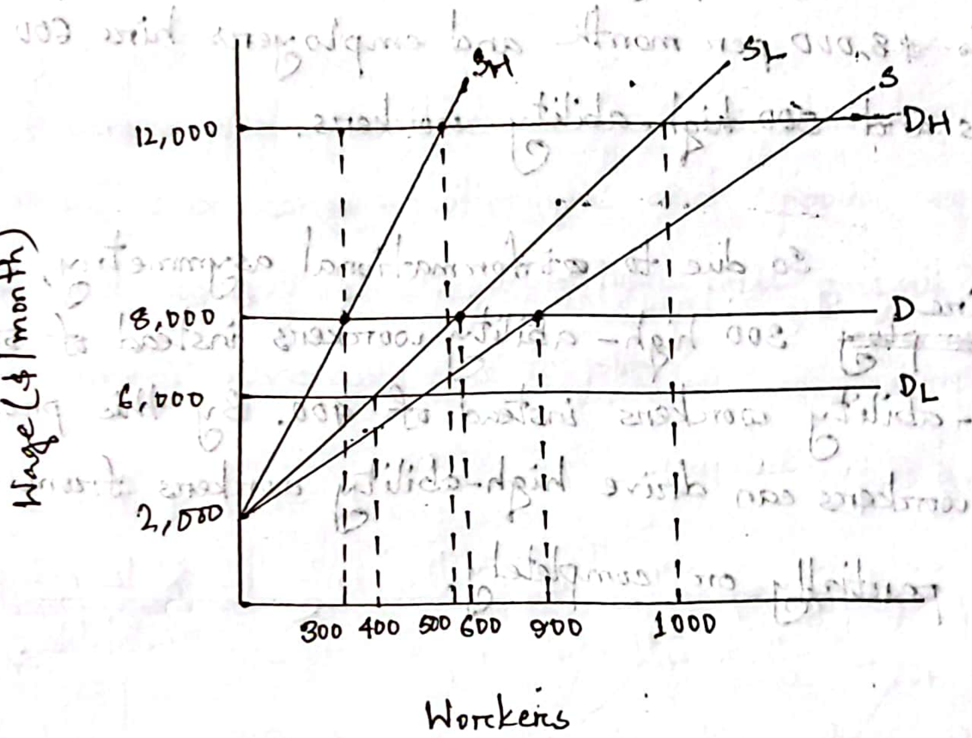


Fig-2: Demand and Supply when ability is unknown

From fig-2, when the wage is $8,000 per month, 600 low-ability workers and 300 high-ability workers are willing to accept employment. When the wage is $12,000, 1,000 low-ability and 500 high-ability workers are willing to accept employment.

Fig-2 also shows the market equilibrium when employers can't observe a worker's ability. Because two-thirds of the available workers have low-ability, an employer should be willing to pay a worker $8,000 per month. The resulting demand curve is labeled D. The curve labeled S is the market supply curve of workers, the horizontal sum of the high- and low-ability supply curve, S_H and S_L. The equilibrium wage is $8,000 per month and employers hire 600 low-ability workers and 300 high-ability workers.

So due to informational asymmetry, employers now hire ~~employ~~ 300 high-ability workers instead of 500 and 600 low-ability workers instead of 400. By this process, low-ability workers can drive high-ability workers from the market either partially or completely.

Ques-13:- How does government response to such informational asymmetries?

Answer:-

When asymmetric information leads to market failures, governments often respond in ways that reduce the potential economic losses. Sometimes the government mandates minimum quality standards, which reduce the asymmetry of information.

Product liability laws also help reduce the effects of asymmetric information. These laws require manufacturers of defective products to compensate buyers for certain types of losses. By imposing prohibitive costs on very low quality firms, they reassure consumers that the available products meet minimum quality standards.

Ques-14:- Considering a simple model of educational attainment distinguish between 'separating equilibrium' and 'pooling equilibrium' with diagram and logic. Which equilibrium will prevail? Why? Does the government have any role to play?

Answer:-

~~Separating Equilibrium:~~ First part is same as "ques-5".

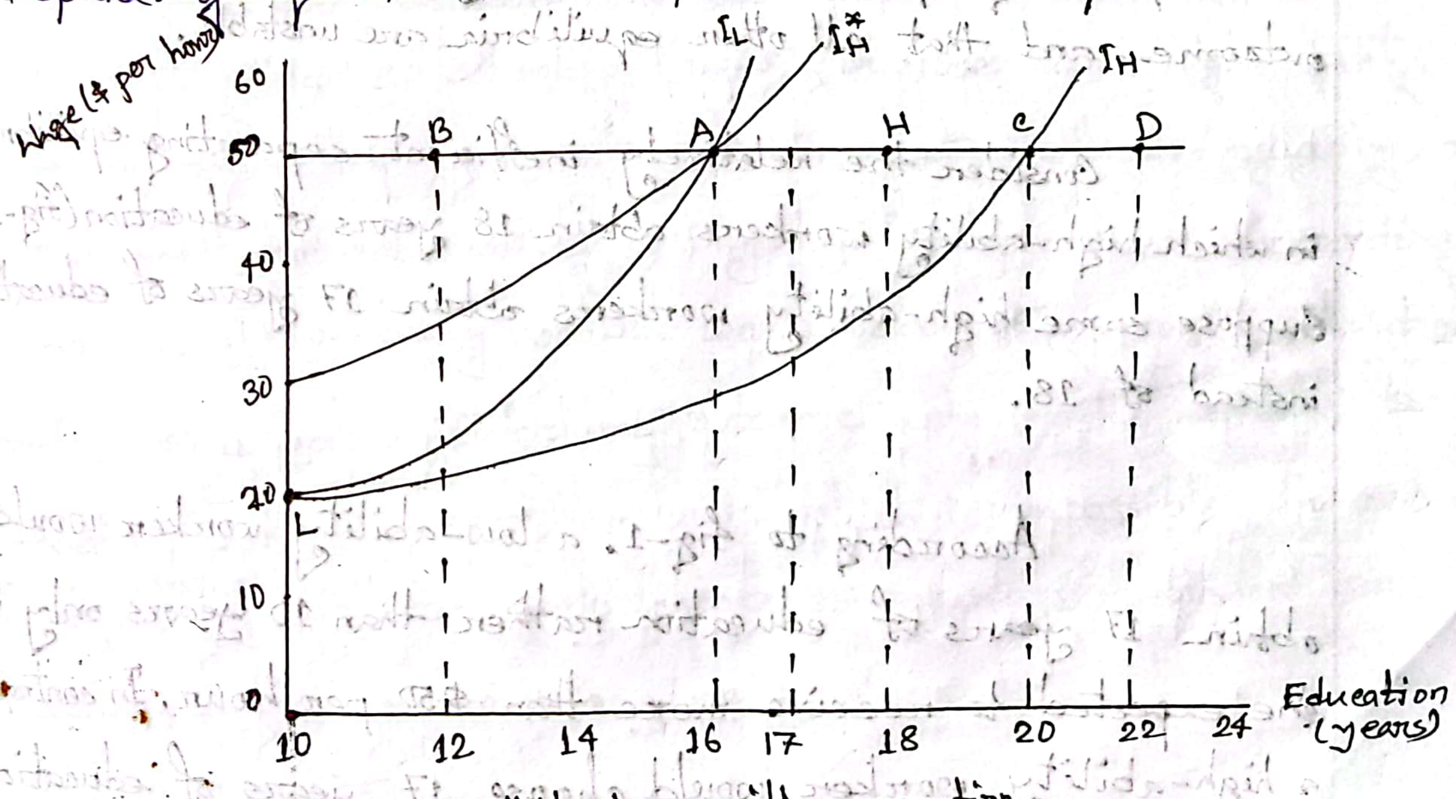


Fig-1: Equilibrium with separation

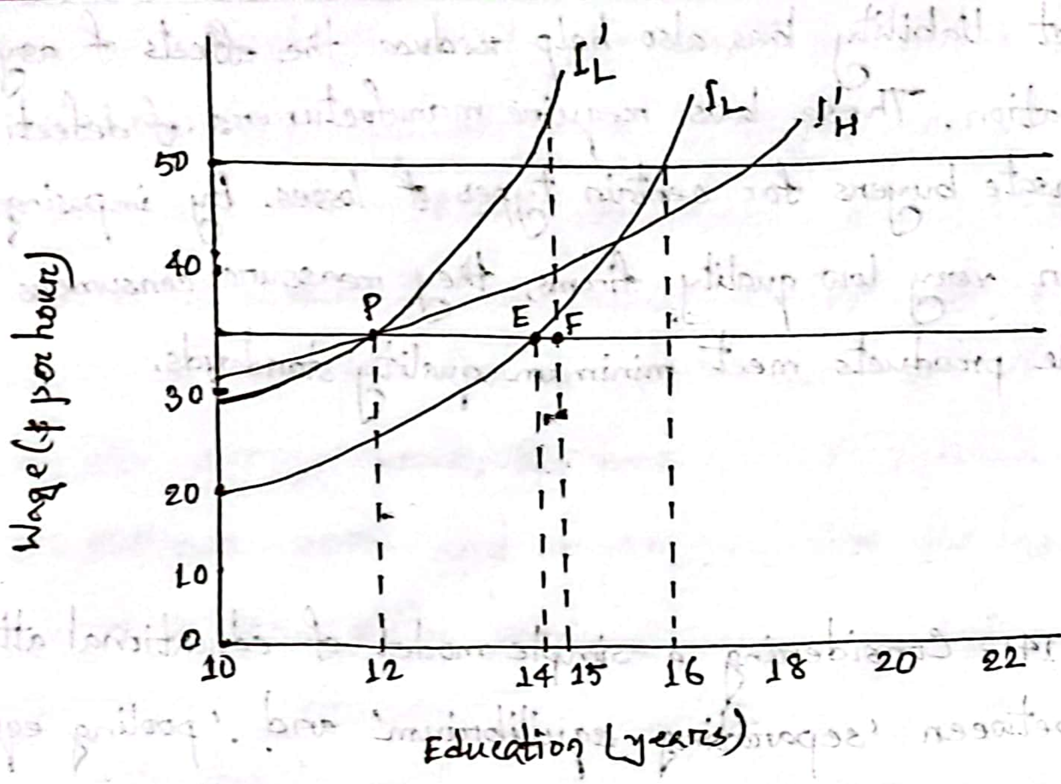


Fig-2: Equilibrium with pooling

Prevailing Equilibrium:-

Some economists think that the most efficient separating equilibrium is the only plausible competitive outcome, and that all other equilibria are unstable.

Consider the relatively inefficient separating equilibrium in which high-ability workers obtain 18 years of education (fig-1). Suppose some high-ability workeres obtain 17 years of education instead of 18.

According to fig-1, a low-ability worker would obtain 17 years of education rather than 10 years only if she expected to receive more than $50 per hour. In contrast, a high-ability worker would choose 17 years of education instead of 18 as long as she expected to receive a wage close to $50 per hour. Therefore, an employer cannot rule out the possibility that the nonconformist has high ability. Putting these inferences together, the employer concludes that the nonconformist must have high ability and is therefore worth $50 per hour.

Because a high-ability worker can anticipate that his competitive wage will be $50 per hour if he gets 17 years of education he has no reason to get 18 years. Sooner or later, some high-ability worker will be willing to take that risk, at which point, the inefficient separating equilibrium will collapse.

For similar reasons, the pooling equilibria may also be unstable. Consider the equilibrium in which all workers obtain 12 years of schooling (fig-2). Suppose some high-ability worker obtains 15 years of education instead of 12. Arguably, an employer should be willing to pay her $50 per hour. According to fig-2, a low-ability worker would obtain 15 years of education rather than her equilibrium outcome only if she expected to receive more than $50 per hour, which is impossible. In contrast, a high-ability worker would choose 15 years of education over her equilibrium outcome as long as she expected to receive a wage close to $50 per hour, which is possible. Therefore, an employer should conclude that the nonconformist must have high ability and is therefore worth $50 per hour. Sooner or later, some high-ability worker will be willing to take that risk, at which point the pooling equilibrium will collapse.

A Possible Role for the Government:-

Because signaling equilibria are often ~~effi~~ inefficient, it's natural to wonder whether some form of government intervention might improve the allocation of resources. If the government were perfectly informed about each worker's ability, it could easily resolve the problems arising from adverse selection.

Suppose that competition tends to produce the most efficient separating equilibrium, but everyone would be better off with the most efficient pooling equilibrium. Government could compel pooling by banning education beyond the fixed level (suppose 10 years).

It also important to remember that, in some cases, signaling can help market participants overcome the problems associated with adverse selection. Banning signaling can then be particularly counterproductive. Without signaling, the market may unravel, driving away all high-ability workers. By preventing a market from unraveling either partially or completely, signaling can therefore enhance social efficiency.

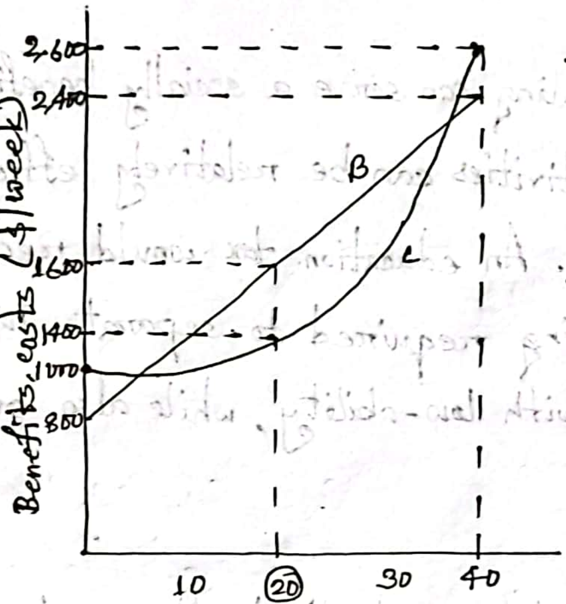
Even though signaling can serve a socially beneficial purpose, taxes on signaling activities can be relatively efficient sources of government revenue. An education tax would reduce the amount of wasteful schooling required to separate high-ability workers from those with low-ability, while also providing resources for public projects.

Question:- "A solution to moral hazard, albiet often an imperfect one" - do you agree with this statement? Why or why not?

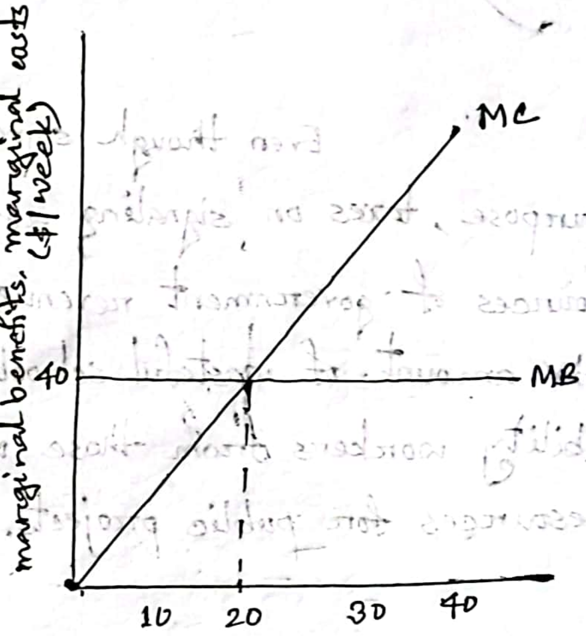
Answer:-

If an employer cannot observe a salesperson's level of effort, she can observe an important consequence of that effort: success in making sales. In such cases, there is a solution to the moral hazard problem, albiet often an imperfect one: provide the trading partner with incentives to take desirable actions by writing a contract or adopting a compensation policy that ties rewards and punishments to observable measures of performance.

Suppose the employer cannot observe the salesperson's level of effort. If the salesperson prefer to relax, he probably won't sell many cars. Unfortunately, the owner can't ~~watch~~ watch him every minute. The salesperson might work harder, though, if the owner structures his compensation so that he ~~earns~~ earns more if he sells more cars.



(a) Benefits and costs



(b) Marginal benefits and marginal costs

Figure-1

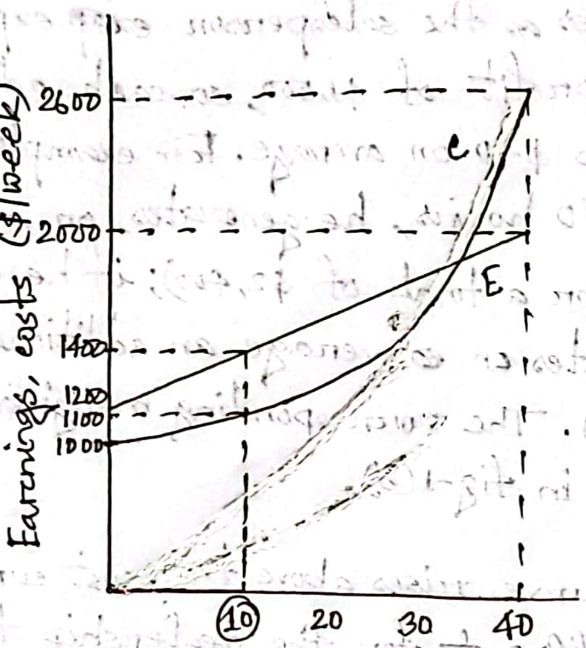
Fig-1(a) illustrates the weekly costs and benefits of the salesperson's decision. The horizontal axis measures his effort in terms of hours per week, (40 hours per week). The curve labeled C shows the relationship between his effort and the personal costs he incurs while working at the dealership. The curve starts at $1000, his opportunity cost of accepting employment. Because effort is costly, the curve is upward sloping. Fig-1(b) shows the corresponding marginal cost curve, labeled MC.

The curve labeled B in fig-1(a) shows the owner's benefit from employing the worker. The curve hits the vertical axis at

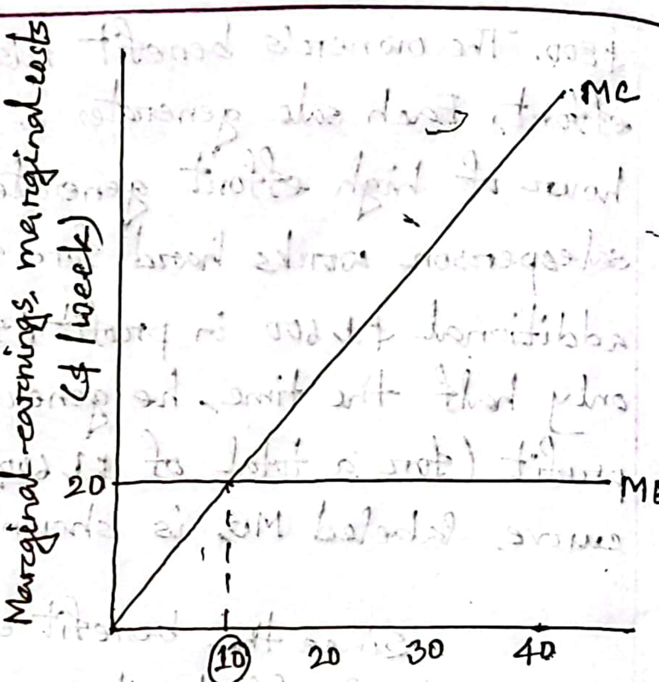
$800. The owner's benefit rises as the salesperson ~~exp~~ expands more effort. Each sale generates a profit of $1,000, so each additional hour of high effort generates $40 on average. For example, if the salesperson works hard for 40 hours, he generates, on average, an additional $1,600 in profit (for a total of $2,400); if he works hard only half the time, he generates on average an additional $800 in profit (for a total of $1,600). The corresponding marginal benefit curve, labeled MB, is shown in fig-1(b).

Since the benefit curve rises above the cost curve for some level of effort, it is efficient for the dealership to employ the salesperson. The efficient outcome entails the salesperson working hard for the number of hours that maximizes the difference between the owner's benefit and the salesperson's cost of effort. As shown in fig-1(b), the solution, 20 hours of hard work each week, equates the marginal benefit and the marginal cost of effort.

As the salesperson's effort is not observable, now, if, instead, the salesperson's compensation is tied to the number of cars he sells, he will have an incentive to work hard. Fig-2(a) illustrates the salesperson's decision if his compensation consists of $1,200 in base pay plus a bonus of $500 for each car he sells. With this incentive scheme, the owner shares the profit from each car ($1000) equally with the salesperson. The curve labeled E reflects the relation between the salesperson's average weekly earnings and his effort. Each hour of effort has a 4 percent chance of generating a sale, so on average it increases



(a) Earnings and costs



(b) Marginal earnings and marginal costs

fig - 2

his pay by $20. So the curve E serves as the salesperson's benefit function. Fig-2(a) also includes his personal cost function. Curves depicting his marginal earnings (ME) and marginal costs (MC) are shown in fig-2(b). The salesperson will choose his effort to maximize the difference between his personal benefits (earnings) and costs. Equivalently, he equates his marginal earnings and marginal costs. As shown in the figure, his best choice is to work hard for 10 hours. With this compensation scheme, his effort remains inefficiently low.

Question:- "Competition between employers can lead to a separating equilibrium only if low-ability workers are sufficiently numerous. If employers cannot observe each other's offers, competition cannot lead to a pooling equilibrium" - do you agree with this statement? Discuss with a hypothetical example.

Answer:-

A Separating Equilibrium:-

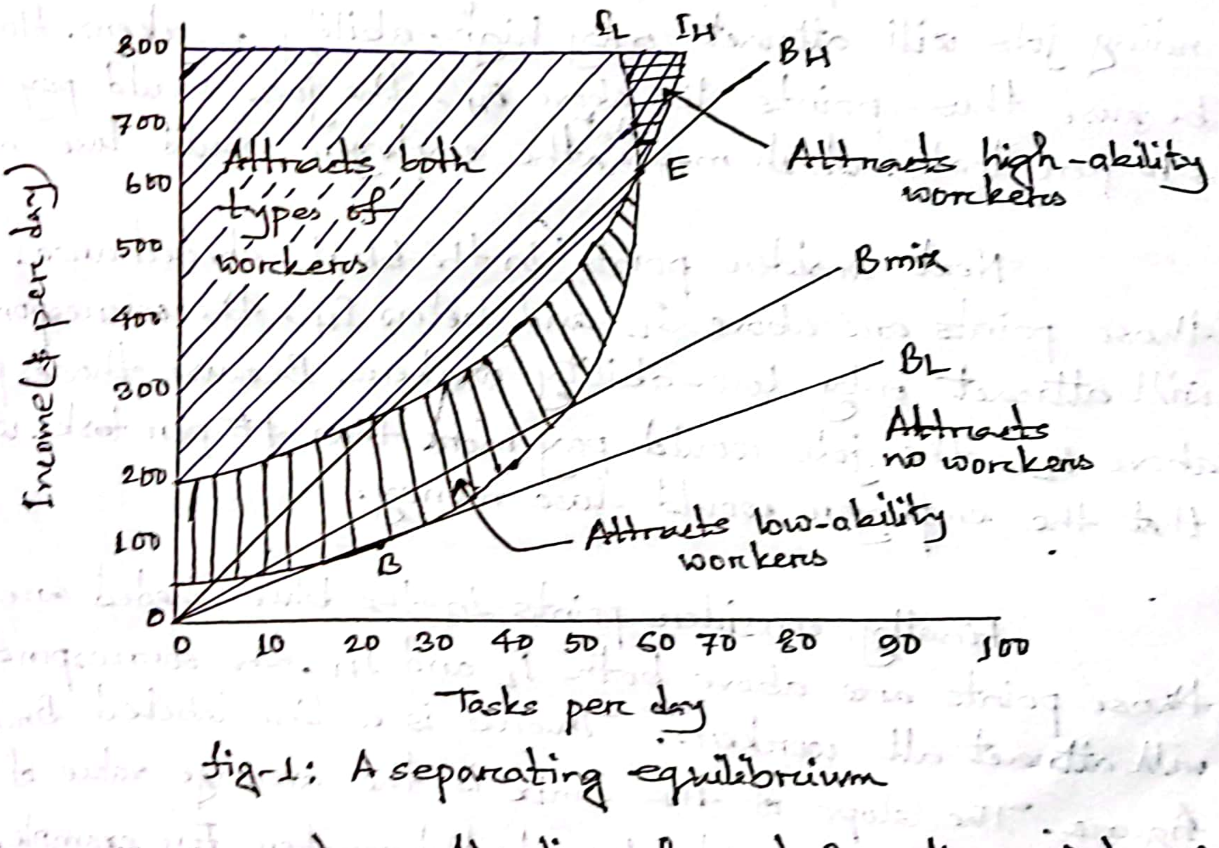


fig-1: A separating equilibrium

Fig-1 produces the lines B_L and B_H, the points B and E, and the indifference curve of a low-ability worker that runs through those points (~~also~~ labeled I_L). We have added the indifference curve for a high-ability worker that runs through point E (labeled I_H). Let's evaluate the profit opportunities available to a new employer. Obviously, jobs associated with points B and E will attract workers, but generate zero profits.

First consider points in the unshaded portion of the figure. Because those points are below both I_H and I_L, the corresponding jobs will not attract any workers. Therefore they aren't profitable.

Next ~~cond~~ consider points in the cross-shaded area. Because those points are above I_H and below I_L, the corresponding jobs will attract only high-ability workers. However, because those points lie above B_H, the jobs would pay more than \$10 per task, which means that the employer would lose money.

Next consider points in the black-shaded area. Because those points are above I_L and below I_H, the corresponding jobs will attract only low-ability workers. Because those points lie above B_L, the jobs would pay more than \$5 per task, which means that the employer would lose money.

Finally, consider points in the blue-shaded area. Because those points are above both I_L and I_H, the corresponding jobs will attract all workers. There is a line labeled Bmix in the figure. The slope of the Bmix is the average value of a task performed by a randomly selected worker. For example, if high- and low-ability workers are equally numerous, then the slope of Bmix is $(1/2 \times \$5) + (1/2 \times \$10) = \$7.50$ per task. A job that attracts all workers isn't profitable unless it lies below Bmix. As the mix of workers shifts from low ability to high ability

B_{mix} rotates upward from B_L to B_H. Therefore, if low-ability workers are sufficiently numerous, B_{mix} passes below the blue-shaded area. In that case, an employer who offered a job corresponding to any point in the blue-shaded area would lose money. However, if high-ability workers were sufficiently numerous, B_{mix} would pass through the blue-shaded area. In that case, points below B_{mix} and above both indifference curves correspond to jobs that would attract both types of workers while generating a profit.

So if low-ability workers are sufficiently numerous, there is a separating equilibrium which employers offer the combination of jobs shown in fig-1 (point B and E). Low-ability workers perform the same number of tasks as high-ability workers and receive the same wages regardless of whether employers know each worker's ability.

Pooling Equilibria:-

In a competitive pooling equilibrium, employers would offer only one type of job, requiring the completion of T_P tasks and paying $\$Y_P$ per day. However, as it turns out there are never any equilibria of this type. To understand why let's consider the following two possibilities.

First, we can rule out the possibility that (T_P, Y_P) is either above or below the line labeled B_{mix} in fig-2. If it were above that line, existing employers would earn positive profits, so new employers would continue to enter the labor market, creating an

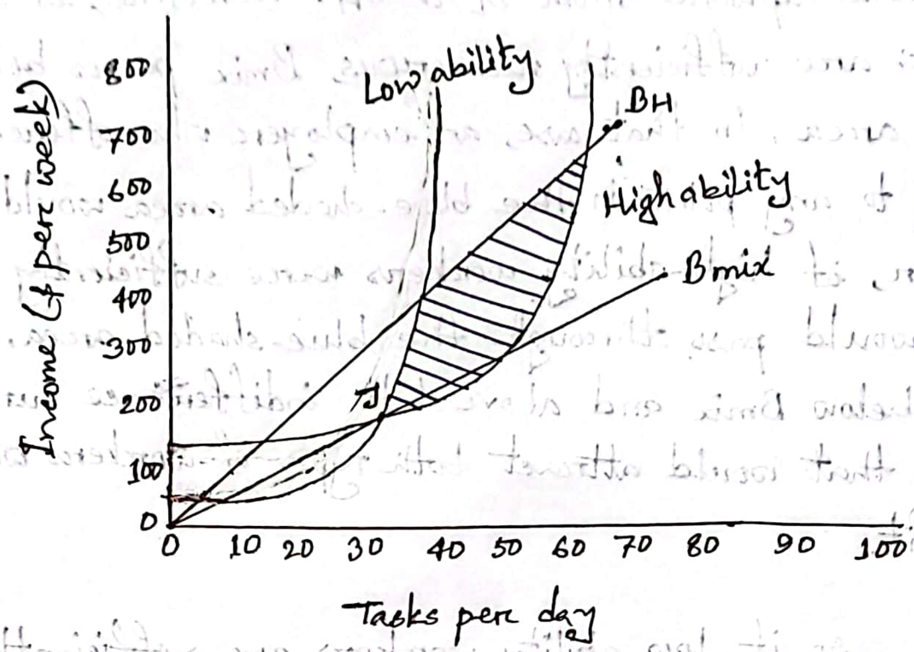


Fig-2:- Pooling does not survive competition

unlimited number of jobs. If it were below the line, employers would turn away applicants to avoid losing money. In either case, the mark wouldn't be in equilibrium.

Second, we can rule out the possibility that (T_P, Y_P) lies on the line B_{mix}. Consider, for example, point J in fig-2. We have drawn two difference curves through point J, one for each type of worker. If an employer created a job with characteristics corres-ponding to any point in the blue-shaded area of the figure (below B_H and the low-ability worker's indifference curve, but above the high-ability workers, while paying them less than $10 per task. Because that strategy permits the employer to earn a profit, the market wouldn't be in equilibrium. Having ruled out every concei-vable possibility, we conclude that there is no pooling equilibrium.

Question:- It is often said that as soon as the buyer of a new car drives it off of a dealer's lot its value fall significantly below the purchase price. Why might this be true? Suppose this pattern held in 1975, but that is no longer true today. What do you think might account for that change? Briefly discuss.

Answer:-

It is often said that as soon as the buyer of a new car drives it off of a dealer's lot, its value falls significantly below the purchase price. One possible reason is that new cars are subject to a higher degree of information asymmetry than used cars. When a buyer purchases a new car from a dealer, they have limited information about the quality and performance of the car. They have to rely on manufacturer's warranty, the dealer's reputation, and their own inspection and test drive. However, these sources of information may not be sufficient or reliable enough to reveal any hidden defects or problems that the car may have. Therefore, the buyer faces a risk of buying lemon, even if it is brand new. This risk reduces the buyer's willingness to pay for the car, and hence lowers its value.

Another possible reason is that new cars are subject to a higher degree of depriciation than used cars. When a buyer purchase a new car from a dealer, they pay a premium price that includes not only the cost of production, but also the dealer's profit margin, taxes, fees, and other charges. However, as soon as the buyer drives the car off the lot, these components of the price are no longer relevant or recoverable. The only thing that matters is the market value of the car, which is determined by its supply and demand. Since new cars are abundant in supply and have many substitutes in the market, their demand is relatively low. Therefore, their market value is significantly lower than their purchase price.

Now if this pattern held in 1975, but no longer true today then one possible explanation is that over time the information asymmetry and depreciation in the new car market has decreased due to various factors. For example:

→ The development of online platforms and databases that provide detailed information about the history and condition of cars has reduced the information gap between buyers and sellers.

→ The improvement of quality standards and regulations in the car industry has reduced the likelihood of manufacturing defects and recalls.

→ The introduction of extended warranties and service contracts by manufacturers and dealers has reduced the risk and cost of repairs for buyers.

→ The increase of consumer awareness and education about car values and depreciation rates has reduced the buyer's overestimation of new car prices.

→ The emergence of new technologies and features (such as hybrid engines, electric vehicles, autonomous driving) that increase the efficiency, performance, and appeal of new cars has increased their demand and value.

This factors have made new cars more transparent, reliable, durable, and desirable than before, thus narrowing the gap between their purchase price and market value now-a-days.

Question:- After reading about signaling and screening, a student named Toha complains: "The world doesn't work this way. Once an employee spends some time at a firm, her employer will learn her true ability and treat her accordingly." Do you agree with him? Why or why not?

Answer:-

I ~~disg~~ disagree with Toha's statement. The world does work this way, at least to some extent. Signaling and screening are not only relevant for the ~~it~~ initial hiring process, but also for the ongoing relationship between the employer and the employee. Here are some reasons why:

→ First, the employer may not be able to observe the employee's true ability even after some time at the firm. For example, the employee may be working in a team, and it may be hard to disentangle her individual contribution from the team's performance. Or the employee may be involved in complex and uncertain tasks, and it may be difficult to measure her output or productivity. In these cases, the employer may still rely on signals, such as education, ~~cert~~ certifications, awards, or recommendations, to infer the employee's ability and potential.

→ Second, the employer may face competition from other firms who are also interested in hiring employee. If the employee has a high ability, she may want to signal it to other potential employers who can offer her better wages and benefits. Similarly, if the employer wants to retain the employee, he may want to screen her ability and offer her incentives that are commensurate with her productivity. This may involve performance evaluations, promotions, bonuses, or other form of feedback

→ Third, the employer and the employee may have dynamic interactions over time that depend on their expectations and beliefs about each other's ability. For example, the employer may invest more in training or mentoring the employee if he believes that she has a high ability and can improve her skills. Or the employee may exert more effort or take more initiative if she believes that the employer values her work and rewards her accordingly. These behaviors can also serve as signals and screens that affect their future relationship.

Therefore, signaling and screening are not only relevant for the initial hiring process, but also for the ongoing relationship between the employer and the employee. They are important tools for reducing asymmetric information and improving efficiency in labore markets.